Indian Stock Market Microstructure Analysis

Project Planning Document

Internship Capstone Project
Team of 3 | 7-Day Sprint Plan
[Date]

1. Project Charter

1.1 Purpose

To analyze historical Indian stock market data (NIFTY 50 / BANK NIFTY / selected NSE stocks) using data analysis, feature engineering, statistical hypothesis testing, and unsupervised/supervised machine learning, in order to understand market behavior and identify distinct market regimes, delivered as a fully documented, reproducible capstone project.

1.2 Objectives

- Deliver a complete, milestone-based Jupyter Notebook covering all 19 steps of the project brief.
- Produce professional supporting documentation: README, requirements.txt, research report, final report, and insights document.
- Demonstrate sound data science practice: reproducibility, clear explanations, honest treatment of limitations.
- Complete the full project within a 7-day sprint using a 3-person team.

1.3 Scope

In Scope

- Daily OHLCV data analysis for one primary index/stock (extensible to more).
- EDA, feature engineering, hypothesis testing, KMeans + Hierarchical clustering, optional supervised regime validation, evaluation.
- Documentation and repository packaging (README, requirements.txt, folder structure).

Out of Scope

- Live trading, backtesting, or trading strategy recommendations.
- Tick-level / order-book microstructure data (cost- and access-prohibitive for this project).
- Production deployment (e.g., a live dashboard) — noted only as future scope.

1.4 Success Criteria

- Notebook runs top-to-bottom without errors on a clean environment using requirements.txt.

Every code cell is preceded by a markdown explanation; every chart has a title, axis labels, and interpretation.

At least one statistically significant or clearly-interpreted null result is reported honestly for the volume–volatility hypothesis.

Clustering produces an interpretable set of regimes validated by silhouette score and (optionally) classifier accuracy.

All final documents (README, research report, final report, insights document) are internally consistent with each other.

2. Team and Roles

Member	Role	Primary Responsibilities
Member A	Data Engineer / EDA Lead	Data acquisition, cleaning, inspection, full exploratory data analysis
Member B	ML & Statistics Lead	Feature engineering, hypothesis testing, clustering, optional supervised learning, evaluation
Member C	Documentation & Insights Lead	Conclusions, README, repo structure, research/final report assembly, presentation

3. Work Breakdown Structure (WBS)

#	Task	Maps to Step(s)	Owner
1	Project statement & microstructure concept	1–2	C (draft), all review
2	Data download via yfinance & column explanation	3–4	A
3	Data loading & inspection	5–6	A
4	Data cleaning (missing values, logic checks, dtypes)	7	A
5	EDA: time series, univariate, bivariate, correlation, volume, returns, volatility, moving averages	8	A
6	Feature engineering (returns, rolling	9	B

	stats, RSI, lags)		
7	Hypothesis testing (T-test, ANOVA, correlation significance)	10	B
8	Clustering: scaling, elbow, silhouette, KMeans, hierarchical	11	B
9	Optional supervised regime validation	12	B
10	Model evaluation (confusion matrix, precision/recall/F1)	13	B
11	Insights & conclusions	14	C
12	README, folder structure, requirements.txt	15–18	C
13	Future scope & improvements	19	C
14	Research report, final report, insights document assembly	—	C, reviewed by A & B

4. Timeline (7-Day Sprint)

Day	Member A	Member B	Member C
1	Kickoff; download data; start cleaning	Kickoff; scaffold feature/ clustering code on sample data	Kickoff; README skeleton; Steps 1–2 draft
2	Finish cleaning; start EDA	Continue scaffolding; integrate real data mid-day	Continue docs; prep notebook templates
3	Finish EDA; hand off to B	Finalize features on real data; run hypothesis tests	Review EDA; draft methodology/limitations
4	Support B; help interpret clusters	Clustering: elbow, silhouette, KMeans, hierarchical	Assemble master notebook as sections land
5	Help write regime interpretations	Optional supervised learning; model evaluation	Fill Insights & Conclusions with real numbers
6	Full notebook re-run & bug fixing	Full notebook re-run & bug fixing	Finalize README, requirements.txt, future scope
7	Final cross-review,	Final cross-review,	Final cross-review,

	rehearse, submit	rehearse, submit	rehearse, submit

5. Risk Register

Risk	Likelihood	Impact	Mitigation
yfinance API/column format changes (e.g., MultiIndex columns, auto_adjust default)	Medium	Medium	Explicitly set auto_adjust=False and flatten MultiIndex columns before saving; validate dtypes immediately after load
Day 3 handoff (A→B) delayed	Medium	High	Hard cutoff on Day 3 regardless of polish level; iterate later if time allows
Statistical tests show non-significant results	Medium	Low	Report honestly; non-significance is a valid, reportable finding, not a failure
.ipynb git merge conflicts	High	Medium	Single-owner-at-a-time editing of the master notebook, or individual section notebooks merged manually in Phase 5
Clustering produces uninterpretable clusters	Low	Medium	Try multiple K values via elbow/silhouette; fall back to fewer, broader regimes if needed
Time overrun on Day 4 (clustering)	Medium	High	Member A shifts to support B immediately if behind schedule

6. Resource Plan

6.1 Tools & Environment

Python 3.10+, Jupyter Notebook

Libraries: pandas, numpy, matplotlib, seaborn, scikit-learn, scipy, yfinance, plotly (see requirements.txt)

Git + shared repository (GitHub)

6.2 Data

Yahoo Finance historical daily data via yfinance (free, no API key required)

Internet access required for the initial download step; data is cached locally as CSV thereafter

7. Communication Plan

Daily 10–15 minute standup (or async chat message): done yesterday / doing today / blockers

Shared document (or repo README) tracking task completion status

Ad-hoc pairing when a handoff is blocked (e.g., A helps B on Day 4 if clustering is behind schedule)

8. Deliverables Checklist

Final Jupyter Notebook (all 19 steps, milestone-structured, fully documented)

README.md

requirements.txt and .gitignore

Research Report (this document's companion)

Final Report

Insights Document

(Optional) Presentation slide deck